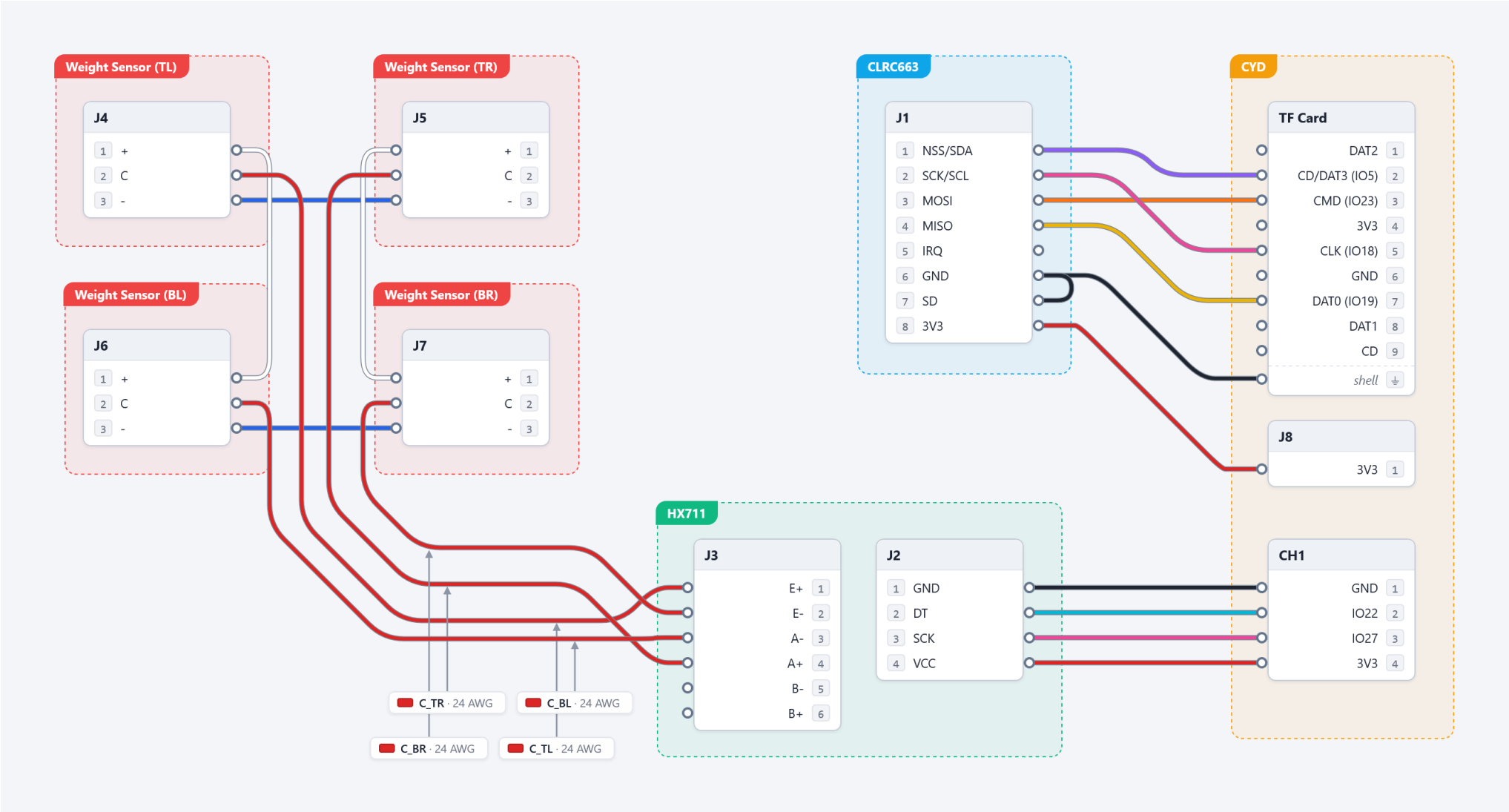
Spoolhouse Scale

Wiring Viz · 2026-07-18 · 19 wires · 10 connectors · 0 cables



Connections

#	Wire	From	To	Color	Gauge	Length	Part	Cable	Net
1		Weight Sensor (TL) › J4.3 (-)	Weight Sensor (TR) › J5.3 (-)	BU	24 AWG				N1
2		Weight Sensor (BL) › J6.3 (-)	Weight Sensor (BR) › J7.3 (-)	BU	24 AWG				N2
3		Weight Sensor (TR) › J5.1 (+)	Weight Sensor (BR) › J7.1 (+)	WH	24 AWG				N3
4		Weight Sensor (TL) › J4.1 (+)	Weight Sensor (BL) › J6.1 (+)	WH	24 AWG				N4
5	C_TL	Weight Sensor (TL) › J4.2 (C)	HX711 › J3.1 (E+)	RD	24 AWG				N5
6	C_TR	Weight Sensor (TR) › J5.2 (C)	HX711 › J3.4 (A+)	RD	24 AWG				N6
7	C_BR	Weight Sensor (BR) › J7.2 (C)	HX711 › J3.2 (E-)	RD	24 AWG				N7
8	C_BL	Weight Sensor (BL) › J6.2 (C)	HX711 › J3.3 (A-)	RD	24 AWG				N8
9		HX711 › J2.1 (GND)	CYD › CH1.1 (GND)	BK	24 AWG				N9
10		HX711 › J2.2 (DT)	CYD › CH1.2 (IO22)	TQ	24 AWG				N10
11		HX711 › J2.3 (SCK)	CYD › CH1.3 (IO27)	PK	24 AWG				N11
12		HX711 › J2.4 (VCC)	CYD › CH1.4 (3V3)	RD	24 AWG				N12
13		CLRC663 › J1.6 (GND)	CYD › TF Card.SHELL	BK	24 AWG				N13
14		CLRC663 › J1.7 (SD)	CLRC663 › J1.6 (GND)	BK/BK	24 AWG				N13
15		CLRC663 › J1.1 (NSS/SDA)	CYD › TF Card.2 (CD/DAT3 (IO5))	VT	24 AWG				N14
16		CLRC663 › J1.3 (MOSI)	CYD › TF Card.3 (CMD (IO23))	OG	24 AWG				N15
17		CLRC663 › J1.2 (SCK/SCL)	CYD › TF Card.5 (CLK (IO18))	PK	24 AWG				N16
18		CLRC663 › J1.4 (MISO)	CYD › TF Card.7 (DAT0 (IO19))	YE	24 AWG				N17
19		CLRC663 › J1.8 (3V3)	CYD › J8.1 (3V3)	RD	24 AWG				N18

Parts & BOM

Part number	Type	Manufacturer	Description	Qty	Cost	Ext. cost
Heat Set Insert M3	Mounting Hardware	Ruthex	https://amzn.eu/d/0b9HPckK	26	0.09	2.34
Countersunk M3x6	Screw			26	0.04	1.04
HX711	PCB		https://amzn.eu/d/09gSECSa	1	1.60	1.60
CLRC663	PCB		https://de.aliexpress.com/item/1005009206202001.html	1	11.07	11.07
Cheap Yellow Display	PCB		https://amzn.eu/d/08TuXSqS	1	17.00	17.00
Kitchen Scale	Ext	Söhnle	https://amzn.eu/d/0i4XcGJX	1	15.00	15.00
Felt Glider 50mm	Ext		https://amzn.eu/d/08yTZHSM	4	0.28	1.12
Misc Wires	Wire			1	1.00	1.00
Bambulab PLA	Filament			0.4	16.00	6.40
					Total	**56.57**

Assembly instructions

Spoolhouse Scale — Assembly

Build the Spoolhouse scale from a donor kitchen scale, an ESP32 Cheap Yellow Display (CYD), an HX711 load-cell amplifier, and a CLRC663 NFC reader, held together by a set of printed parts.

What you need

Printed parts — PLA, PETG, or any other material:

Part	Quantity
Sensor foot	4
Sensor base	4
MainPlate	1
MainboardHousing	1
CoverFront / CoverBack	1 each

> Tip: print the sensor feet/bases and the two covers in a contrasting color.

Hardware & electronics

- Donor kitchen scale (provides the 4 weight sensors and their mounting clips)
- ESP32 Cheap Yellow Display (ESP32-2432S028R)
- HX711 load-cell amplifier
- CLRC663 NFC reader
- 26 heat-set inserts
- 26 M3x6 countersunk screws (16 for the sensors, 6 for the covers, 4 for the CYD)
- 4 felt pads, 50 mm

Tools: soldering iron, hot-air gun (recommended for desoldering), small flat screwdriver, glue.

1. Print the parts

Print everything in the table above. Sensor foot and sensor base are separate parts so the print supports can be removed.

Press the 26 heat-set inserts into the holes in the MainPlate and the MainboardHousing.

2. Strip the kitchen scale & fit the sensor feet

1. Remove the rubber pads from the bottom of the scale and unhinge the clips with a small screwdriver. Keep the clips — they are reused later.
2. Open the scale housing and slide the weight sensors out of the upper housing.
3. Unsolder the sensor wires from the scale's mainboard.
4. Slide each weight sensor into a sensor base and attach the bases to the MainPlate with the 16 M3x6 countersunk screws.
5. Clip a sensor foot onto each sensor using the clips saved in step 1.
6. Stick a 50 mm felt pad onto each sensor foot — it covers the clip slots and keeps the scale from scratching soft surfaces.

3. Prepare the Cheap Yellow Display

Desolder the TF-card slot and the connectors marked P3 and CN1 (preferably with a hot-air gun). The sensor wires

are soldered to these pads. Mount the CYD with 4 M3x6 countersunk screws to the MainboardHousing.

4. Wiring

1. Glue the HX711 into the center of the MainPlate.
2. Place the CLRC663 on the MainPlate and melt the printed guide over the reader with the soldering iron so it is locked in place.
3. Solder all connections according to the wiring diagram.

5. Finishing touches

Fasten the covers with the 6 M3x6 countersunk screws — this simultaneously clamps the MainboardHousing to the MainPlate.

Build and flash the firmware over the USB-C connector (only the first flash needs a cable — later updates go over WiFi, see the manual below).

Spoolhouse Scale — Manual

A WiFi filament scale built on an ESP32 Cheap Yellow Display (CYD). Put a spool on it, and it reads the weight and — if the spool has an NFC tag — its filament details, then uploads the reading to your Spoolhouse server with one tap.

- Display: current weight (zeroable) + decoded filament info
- NFC: reads Bambu Lab (and, experimentally, Creality / OpenTag3D) spool tags
- Upload: sends the measured weight to Spoolhouse over WiFi (registers or refreshes the spool by its tag)
- Config: a built-in web page for setup, calibration, and firmware updates — no cable needed after first flash

1. Quick start

1. Power it on (USB-C). On first boot with no saved WiFi it opens a setup Access Point — see First-time setup.
2. Once connected, the display shows the weight and "tap screen to tare".
3. Place a spool. If its tag is recognized, its material/colour/details appear and the hint changes to "tap screen to upload".
4. Tap the screen. It measures for 5 seconds, then shows "Uploaded!" — the weight is now in Spoolhouse.
5. Manage everything from a browser at http://spoolhouse-scale.local/ (see The config page).

2. First-time setup (WiFi)

If the scale can't reach a known WiFi network, it starts a setup portal:

1. The screen shows "WiFi setup — join `Spoolhouse-Setup`, open 192.168.4.1".
2. On your phone/laptop, join the `Spoolhouse-Setup` WiFi network.
3. A captive portal opens (or browse to http://192.168.4.1). Choose:
   - your WiFi network and enter its password,
   - the Spoolhouse API token (`sph_…`, from the web app's Account -> API tokens),
   - the Spoolhouse base URL (e.g. `https://spoolhouse.example.com`).
4. Save. The device reboots and connects. Done.

> Only 2.4 GHz WiFi is supported (the ESP32 has no 5 GHz radio).

After setup you can change the token / base URL any time from the config page — the portal only appears when WiFi is down.

3. Everyday use

Weighing & uploading

- Empty scale reads 0 g (it tares on boot).
- Tare (zero): when no recognized spool is present, tap the screen (or press the on-board BOOT button). The footer shows "tap screen to tare". The device waits 1 second for the tap's wobble to settle, then measures for 2 seconds (averaging like an upload, see below) — keep hands off until it shows "Tared".
- Weigh + upload: place a tagged spool -> its details appear and the footer shows "tap screen to upload" -> tap the screen. The device:
    1. waits 1 second for the tap's wobble to settle,
    2. measures the weight for 4 seconds,
    3. averages the middle of the readings (discarding the noisy 5 % extremes),
    4. uploads that weight to Spoolhouse, and
    5. shows "Uploaded!" with the value.

The upload calls `POST /api/v1/spools/weigh`, which registers the spool on first sight (creating the material/spool-type from the tag) or refreshes its weight thereafter. New spools register with an assumed 250 g empty-spool weight — adjust the spool type in the web app for exact "filament remaining".

What the display shows

- Top: title + WiFi indicator.
- Middle: spool info when a tag is recognized — manufacturer, material, a colour swatch, nominal weight / diameter / length, spool ID, production date.
- If a tag isn't recognized: "Unknown tag" + its UID.
- Bottom: the large weight readout, then the tap hint.

Supported filament tags

- Bambu Lab — fully supported and verified (MIFARE Classic; keys derived from the tag UID).
- OpenTag3D (opentag3d.info) — the device both reads and writes this open format (NDEF on NTAG21x). This is what writing a tag produces.
- Creality and OpenPrintTag — decoders are implemented but not yet verified on real tags; treat as experimental.

3.5 Writing a tag for a spool without one

Give a "dumb" (tag-less) spool its own NFC tag so the scale recognises it like a smart spool. You need a blank NFC tag (NTAG21x — see the note below).

1. Open the config page -> Write custom tag.
2. Fill in manufacturer, material name, filament type (e.g. PLA), colour, nominal filament weight, and empty-spool weight.
3. Press Write tag. The device screen shows "Hold blank tag to the reader".
4. Hold the blank tag against the NFC reader. It writes an OpenTag3D record and shows "Tag written"; the page confirms too. (Press cancel on the page to abort.)
5. Stick the tag on the spool. From now on, placing it on the scale identifies it and uploads its weight like any smart spool — its NFC UID becomes the spool's unique id.

> Tag type: any NTAG213/215/216 works. The empty-spool weight lives at a high offset in the OpenTag3D layout that only fits NTAG215/216; on a smaller NTAG213 everything else is written and the empty weight falls back to the 250 g default (adjust the spool type in the web app). MIFARE Classic tags can't be written.

4. Calibrating the weight

Do this once (or whenever readings drift). Easiest via the config page:

1. Remove everything from the scale, press Tare (zero).
2. Place a known weight (e.g. a 500 g calibration weight).
3. Enter that weight in grams and press Calibrate.

The device computes and stores the calibration factor (counts/gram) and applies it immediately — no reflash. It's kept in flash and survives reboots (a factory reset returns it to the firmware default).

5. The config page

Open http://spoolhouse-scale.local/ (or the device's IP, e.g. `http://192.168.1.184/`) from any device on the same network.

> Use http://, not `https://` — the device serves plain HTTP. The page has no login (it's meant for your trusted LAN). Form submissions are protected against cross-site requests.

Sections:

- Weight — live reading + Tare (zero).
- Calibrate — guided calibration (see above).
- Spoolhouse server — change the base URL and API token (leave the token blank to keep the current one).
- Write custom tag — encode a blank NFC tag for a tag-less spool (see 3.5).
- Display — Reset display re-initialises the screen if it has gone blank/white (see the troubleshooting note); the device keeps running on WiFi meanwhile.
- Firmware update — upload a new `.bin` (see below).

The page follows the Spoolhouse web-app design and your phone's light/dark theme.

6. Updating the firmware

Three ways, no case-opening needed after the first flash:

A. Browser upload (easiest)

1. Build the firmware (`pio run -e cyd`) — the file is `hardware/.pio/build/cyd/firmware.bin`.
2. On the config page -> Firmware update -> choose the `.bin` -> Upload & flash.
3. The screen shows progress; the device reboots into the new firmware.

B. Over-the-air from PlatformIO

```
pio run -e cyd_ota -t upload --upload-port spoolhouse-scale.local
# or use the device IP; add the OTA password if set:
#   --upload-flags "--auth=<OTA_PASSWORD>"
```

> On Windows you may need to allow the PlatformIO Python through the firewall the first time (it receives the image transfer).

C. USB

```
pio run -e cyd -t upload
```

If a bad/incompatible image is uploaded, the update fails safely and the device keeps running the current firmware (it only reboots on success).

7. Factory reset

Erases the saved WiFi, API token, base URL, and calibration (reverting to the compiled defaults) and reopens the setup portal.

1. Press and hold the touchscreen.
2. At 10 s a countdown warning appears; at 15 s it turns red. Release any time before 20 s to cancel.

3. At 20 s a confirmation screen shows Go back and RESET.
4. Lift your finger, then deliberately tap RESET to wipe. Tapping anywhere else (or waiting) cancels — so an accidental hold can't reset the device.

After a reset the scale boots into the WiFi setup portal.

8. Hardware & wiring

Board: ESP32-2432S028R "Cheap Yellow Display" (ILI9341 TFT + XPT2046 touch).

Peripheral	Signal	ESP32 pin
HX711 load cell (5 kg)	DT	IO22
	SCK	IO27
PN532 NFC (SPI mode)	MOSI	IO23
	MISO	IO19
	SCK	IO18
	SS	IO5 (former SD-card CS pad; SD holder removed)
XPT2046 touch (on-board)	CLK / CS / MOSI / MISO / IRQ	IO25 / IO33 / IO32 / IO39 / IO36

Notes:

- The TFT uses the CYD's stock wiring on the HSPI peripheral. Touch uses the VSPI peripheral. The PN532 is bit-banged software SPI on its own pins, and runs on a separate CPU core so NFC reads don't stutter the display.
- The PN532's SS must go to a real GPIO (IO5), not ground — it wakes on a falling SS edge and stays silent if SS is tied low.

9. Troubleshooting

Symptom	Cause / fix
spoolhouse-scale.local won't load	Use http://, not https. If mDNS .local doesn't resolve, use the device's IP (shown by your router, or on the serial log as [wifi] connected: <ip>).
Setup AP appears unexpectedly	The saved WiFi wasn't reachable. Re-enter it in the portal.
NFC not detected / "NFC reader offline"	Check the PN532 wiring, especially SS -> IO5 (not GND) and MOSI/MISO not swapped. The reader re-initialises every few seconds.
Tag shows "Unknown tag"	Not a supported tag (only Bambu Lab is verified), or a transient read — re-seat the spool.
Colours look wrong on the display	The panel runs inverted + a gamma tweak is applied in firmware; if a swatch still looks off, it's the tag's actual (often muted) colour.
Weight is negative or wildly off	Re-calibrate (Section 4). The load-cell polarity sets the sign automatically during calibration.
Screen goes white but the device still responds on WiFi	The ILI9341 has no wired reset line, so a brief power/SPI glitch — most often when the NFC reader energises during a tag read on a marginal 5 V supply — can blank it while the ESP32 keeps running. Hit Reset display on the config page to recover without a reboot (or power-cycle). Mitigations in firmware: a conservative 40 MHz TFT clock, and the display repaint is held off while the NFC bus is actively reading so the two don't overlap. A solid 5 V supply / good cable is the real cure (the CYD's USB VCC is often marginal).
Screen shows noise on boot, then whites out and the device drops off WiFi too	A full brownout (thin/long USB VCC line) resetting the whole board when WiFi powers up. Use a solid 5 V source / good cable; the serial reset reason RTCWDT_BROWN_OUT_RESET confirms it.
Serial port keeps dropping when flashing	Flaky USB data lines — use OTA/browser updates instead, or a known-good cable/port.

10. Building from source

Requires PlatformIO.

1. Copy `hardware/include/secrets.h.example` -> `hardware/include/secrets.h` and fill in WiFi (optional fallback), `SPOOLHOUSE_URL`, `SPOOLHOUSE_TOKEN`, and `OTA_PASSWORD`. `secrets.h` is git-ignored.
2. Build & flash:

```
pio run -e cyd -t upload        # USB
pio device monitor              # serial log @ 115200
```

3. Partition scheme is `min_spiffs` (two OTA app slots) so over-the-air / browser updates work — don't switch to `huge_app`, which has a single app slot and can't OTA.

The source is under `hardware/src/` (`main.cpp`, `scale.*`, `nfc/`, `spool/`, `net/`, `config_store.*`). Runtime config (token, URL, calibration, WiFi) lives in NVS; the `secrets.h` values are only compile-time defaults / fallbacks.